ZERO TRUST SECURITY FRAMEWORK

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Tools Used: Zero Trust Security Concepts

Platform: Enterprise Security Architecture

Introduction

Traditional security models rely on the concept of trusting users and devices within the internal network. However, modern cyber threats require a stronger security approach. Zero Trust is a cybersecurity framework that assumes no user or device should be trusted automatically. Every access request must be verified before granting access to resources.

Objective

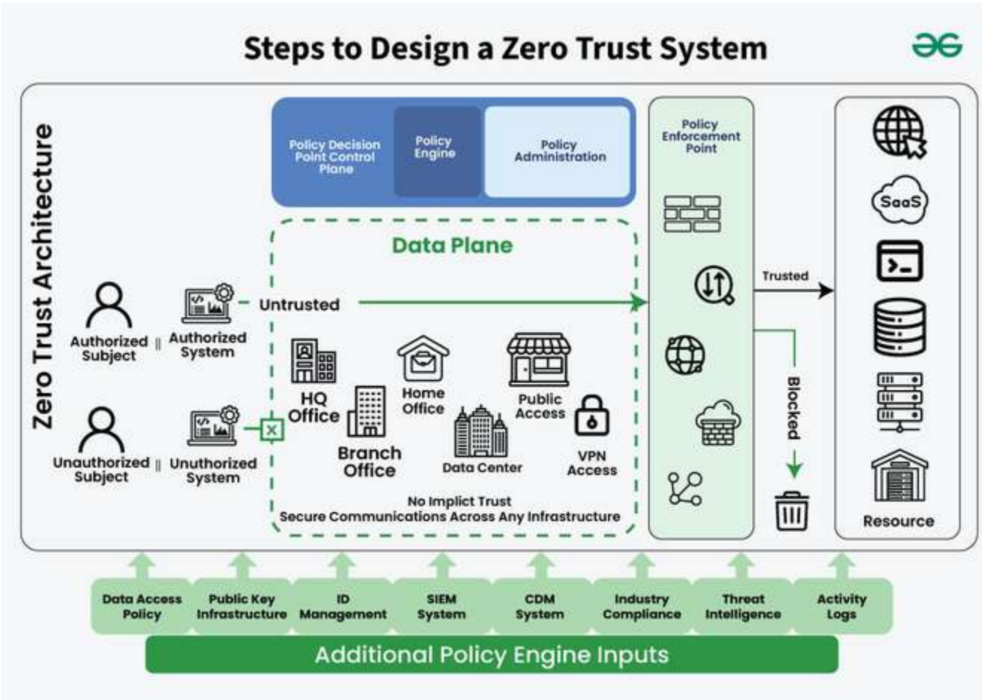
The objective of this project is to understand the principles of the Zero Trust security model and how it helps organizations protect sensitive resources from unauthorized access.

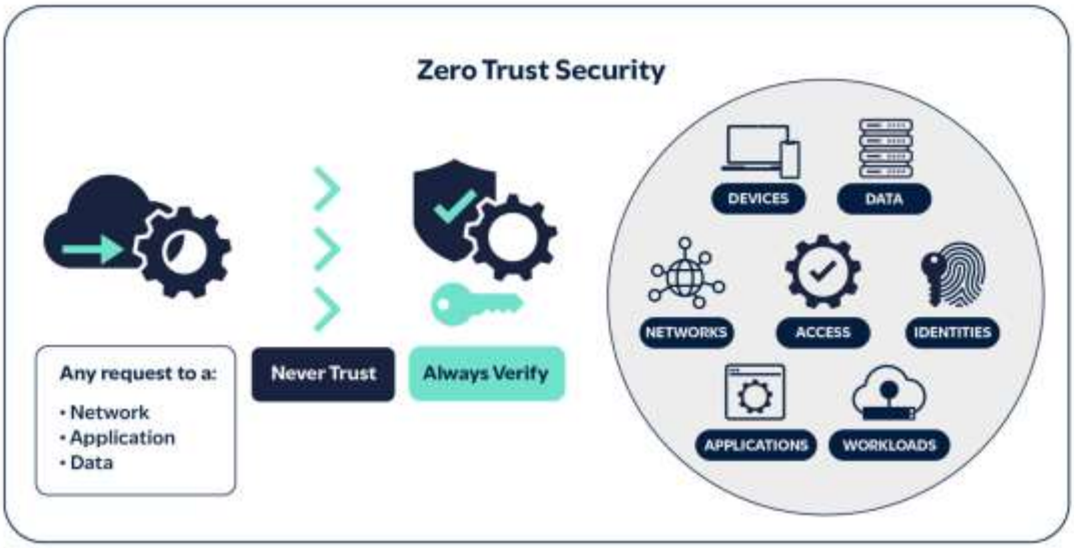
Tools Used

Zero Trust architectures use technologies such as identity verification systems, multi-factor authentication, network segmentation, and continuous monitoring tools.

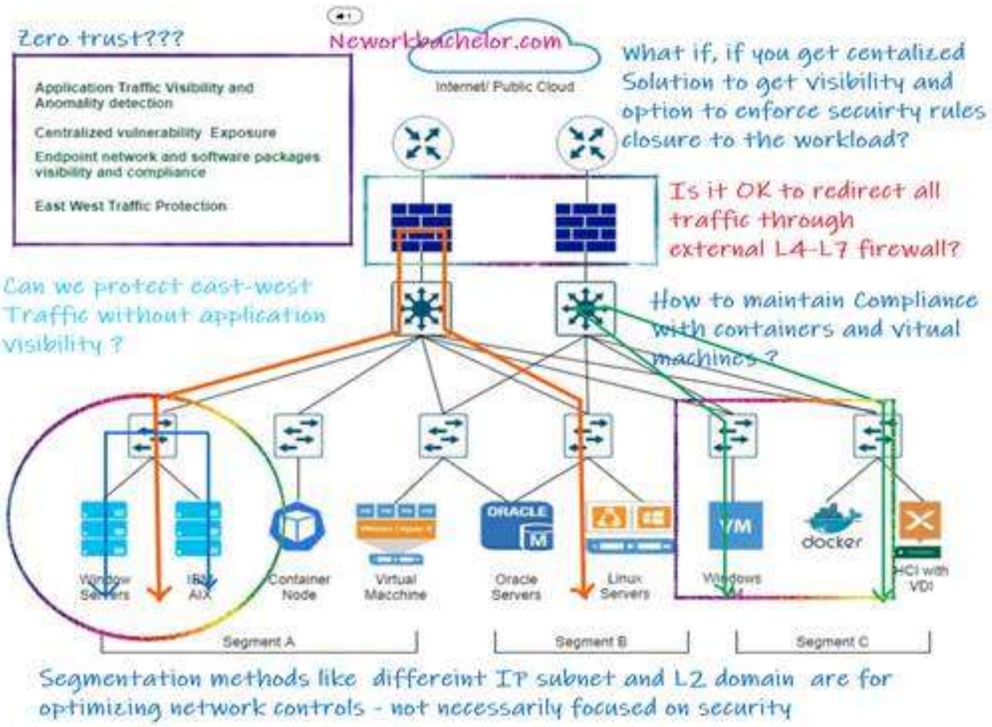
Methodology

The Zero Trust model verifies users and devices before granting access to resources. Authentication and authorization mechanisms ensure that only legitimate users can access systems. Continuous monitoring helps detect suspicious activities within the network.





Zero-Trust Security with Microsegmentation



Results

The Zero Trust framework improves security by enforcing strict access control policies and continuous verification of users and devices.

Security Analysis

Implementing Zero Trust reduces the risk of insider threats and unauthorized access. Even if attackers gain access to a network, they cannot move freely without continuous authentication and verification.

Conclusion

Zero Trust security frameworks provide a modern approach to cybersecurity by enforcing strict identity verification and access control. This model helps organizations protect critical systems and data from cyber threats.